

Caleb O’Neal

(571) 249-9134 | caleboneal07@gmail.com | github.com/caleboneal | caleboneal.dev

EDUCATION

Bachelor of Science in Computer Engineering University of Virginia, School of Engineering and Applied Science	Expected May 2028 <i>Cumulative GPA: 3.94/4.00</i>
Associate of Science in Computer Science Northern Virginia Community College	June 2025 <i>Cumulative GPA: 3.84/4.00</i>

WORK EXPERIENCE

Intern Virginia Tech Qualcomm Thinkabit Lab	August 2024 – May 2025 <i>Falls Church, VA</i>
• Built a web-based simulation of a BLDC motor to demonstrate motor-control fundamentals using React and Rust • Developed curriculum materials and led STEM workshops for visiting K–12 student groups	
Teaching Assistant Arlington Career Center	January 2025 – May 2025 <i>Arlington, VA</i>
• Instructed 15 students in a college-level statistics course covering statistical analysis and real-world data interpretation • Evaluated assignments and research posters, critiquing students’ analysis of relationships in real-world datasets and work on topics including ANOVA tests, regression analysis, and data visualization	
Intern United States Government	June 2024 – August 2024 <i>Virginia</i>
• Redesigned a department website to guide agency management to enterprise architecture data visualization tools • Migrated the 20+ Git repositories between Git hosts, enabling new CI/CD features including automated testing and deployment and the ability to work remotely on non-classified code • Collaborated with the team’s Scrum Master to follow Agile principles including regular code reviews and sprint planning	

PROJECTS

Occupancy Tracker, Team of 5	September 2025 – December 2025
• Programmed and built an Arduino-based system to track real-time occupancy in a room by detecting entry/exit through a doorway using a pair of ultrasonic sensors connected to an LCD display listing current occupancy • Implemented direction detection logic to count entries/exits using moving average filtering and debouncing techniques • Maintained accurate counts across more than 30 test crossings	
FIRST Robotics Competition Team, Team Captain, Programming Lead	August 2022 – May 2025
• Captained a 50-member robotics team with a \$50,000 annual budget to deliver consistently high performing robots • Redesigned and formalized the team management structure following significant membership growth • Wrote Java code to control autonomous and teleoperated robot behavior including autonomous navigation and balancing on field elements that led to a peak autonomous ranking of 27th of approximately 3,300 teams • Wrote robot controls logic using PID controllers that incorporated data from computer vision systems, IMUs, and rotary encoders using a Kalman filter	
NASA App Development Challenge, Team of 6	October 2023 – May 2024
• Developed a 3D interactive simulation of lunar surface exploration for a potential landing site for NASA Artemis III mission in Python, simulating landing, navigating on the lunar surface, and communication with the Lunar Communications Relay • Selected to present our simulation to software professionals at the PyCon US 2024 poster presentation session	

SKILLS

- **Programming:** Python, C, C++, Java, JavaScript, Rust, Zig, React
- **Tools:** Git, CI/CD, Linux, KiCad, QEMU